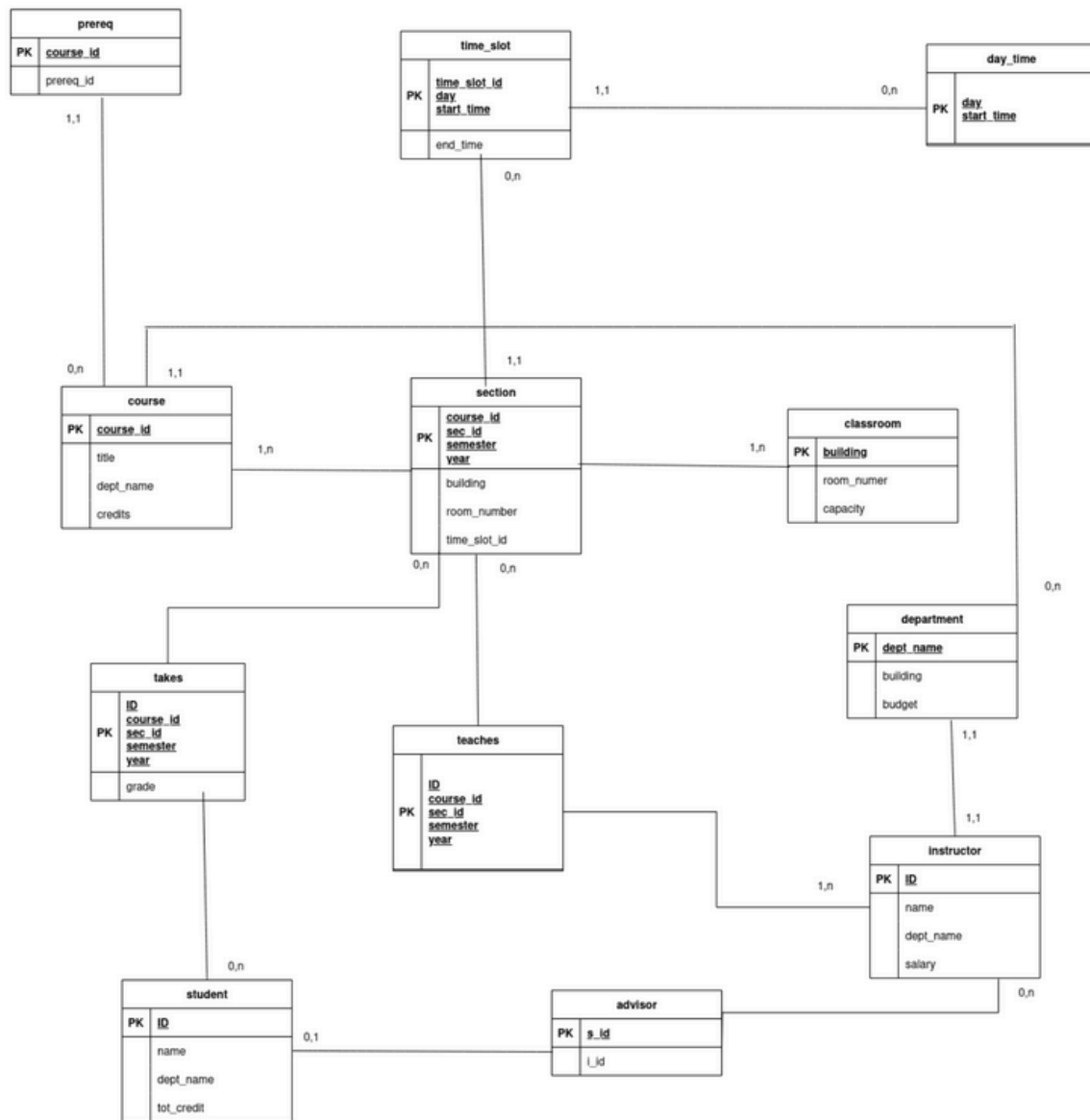


Compte Rendu TP n° 1 BD

Exercice 1:

1.1 Diagramme Entité-Association



Exercice 2 :

1. **SELECT * FROM** section **NATURAL JOIN** course;
2. **SELECT * FROM** course;
3. **SELECT** title, dept_name **FROM** course;

4. **SELECT** dept_name, budget **FROM** department;
5. **SELECT** name, dept_name **FROM** teacher;
6. **SELECT** name **FROM** teacher **WHERE** salary > 65000;
7. **SELECT** name **FROM** teacher **WHERE** salary **BETWEEN** 55000 AND 85000;
8. **SELECT DISTINCT** dept_name **FROM** teacher;
9. **SELECT** name **FROM** teacher **WHERE** salary > 65000 AND dept_name = 'Comp. Sci.';
10. **SELECT * FROM** section **WHERE** semester = 'Spring' **AND** year = 2010;
11. **SELECT** title **FROM** course **WHERE** credits > 3 **AND** dept_name = 'Comp. Sci.';
12. **SELECT** name, dept_name, building **FROM** teacher **NATURAL JOIN** department;
13. **SELECT DISTINCT** student.*
FROM takes, student, course
WHERE takes.ID = student.ID
AND takes.course_id = course.course_id
AND course.dept_name = 'Comp. Sci.';
14. **SELECT DISTINCT** student.name
FROM takes, student, course, teacher, teaches
WHERE takes.ID = student.ID
AND takes.course_id = course.course_id
AND teacher.name = 'Einstein'
AND teaches.course_id = takes.course_id
AND teaches.ID = teacher.ID;
15. **SELECT DISTINCT** course.course_id, teacher.name
FROM (teaches **NATURAL JOIN** course) **NATURAL JOIN** teacher;
16. **SELECT DISTINCT** course_id, name
FROM (teaches **NATURAL JOIN** course) **NATURAL JOIN** teacher;
17. **SELECT** course_id, count(ID)
FROM ((takes **NATURAL JOIN** student) **NATURAL JOIN** student)
WHERE semester = 'Spring' **AND** year = 2010
GROUP BY course_id;
18. **SELECT** dept_name, max(salary)
FROM (department **NATURAL JOIN** teacher)
GROUP BY dept_name;
19. **SELECT** course_id, count(ID)
FROM (takes **NATURAL JOIN** student)
GROUP BY course_id
ORDER BY count(ID) **DESC**;
20. **SELECT** building, count(course_id)
FROM section
WHERE (semester = 'Autumn' **AND** year = 2010)
OR (semester = 'Spring' **AND** year = 2010)
GROUP BY building;
21. **SELECT** dept_name, count(course_id)
FROM Department **NATURAL JOIN** course
GROUP BY dept_name;

22. **SELECT** dept_name, count(course_id)
 FROM (((section **NATURAL JOIN** course) **NATURAL JOIN** teaches) **NATURAL JOIN** teacher) **NATURAL JOIN** department
 GROUP BY dept_name;

23. **SELECT DISTINCT** title, name
 FROM ((section **NATURAL JOIN** teaches) **NATURAL JOIN** teacher) **NATURAL JOIN** course;

24. **SELECT** semester, **count**(course_id)
 FROM section
 GROUP BY semester;

25. **SELECT** student.name, sum(credits)
 FROM (student **NATURAL JOIN** takes), course
 WHERE course.course_id = takes.course_id
 AND student.dept_name <> course.dept_name
 GROUP BY id, name;